A RATIONAL MAP FROM A REDUCED ALGEBRAIC SPACE TO A SEPARATED ONE IS REPRESENTED BY A MORPHISM ON ITS DOMAIN OF DEFINITION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Uniqueness of the representing morphism is part of the statement; the proof establishes it by showing any two representatives agree on the whole intersection, which is exactly what makes the glued morphism unique.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Representatives of a rational map agree on a dense open. used at proof lines 9821-9822. Supplies the dense open W on which the two representatives coincide.
- The diagonal of a separated morphism is a closed immersion, hence equalizers are closed. used at proof lines 9823-9826. Makes the equalizer E a closed subspace of V intersect V'.
- A morphism whose target is reduced is determined by its underlying map of topological spaces. Properties of Spaces, Lemma spaces-properties-lemma-map-into-reduction, used at proof lines 9827-9830. Upgrades |E| = |V intersect V'| to an equality of subspaces.
- Glueing morphisms along a surjection of fppf sheaves. used at proof lines 9831-9834. Produces the global morphism f : U -> Y from the matching local representatives.

Source reference. Original lemma 03JJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4641–4641. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let X and Y be algebraic spaces over S. Assume X is reduced and Y is separated over S. Let φ be a rational map from X to Y with domain of definition $U \subset X$. Then there exists a unique morphism $f : U \to Y$ of algebraic spaces representing φ.*

Proof. Let (V, g) and (V', g') be representatives of φ. Then g, g' agree on a dense open subspace $W \subset V \cap V'$. On the other hand, the equalizer E of $g|_{V \cap V'}$ and $g'|_{V \cap V'}$ is a closed subspace of $V \cap V'$ because it is the base change of $\Delta : Y \to$

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$Y \times_S Y$ by the morphism $V \cap V' \to Y \times_S Y$ given by $g|_{V \cap V'}$ and $g'|_{V \cap V'}$. Now $W \subset E$ implies that $|E| = |V \cap V'|$. As $V \cap V'$ is reduced we conclude $E = V \cap V'$ scheme theoretically, i.e., $g|_{V \cap V'} = g'|_{V \cap V'}$, see Properties of Spaces, Lemma 03JJ. It follows that we can glue the representatives $g : V \to Y$ of φ to a morphism $f : U \to Y$ because $\coprod V \to U$ is a surjection of fppf sheaves and $\coprod_{V,V'} V \cap V' = (\coprod V) \times_U (\coprod V)$. $\square$